

CONTRIBUTION OF STRENGTH CHARACTERISTICS TO CHANGE OF DIRECTION AND AGILITY PERFORMANCE IN FEMALE BASKETBALL ATHLETES

TANIA SPITERI,^{1,2} SOPHIA NIMPHIUS,^{1,2} NICOLAS H. HART,^{1,3} CHRISTINA SPECOS,²
JEREMY M. SHEPPARD,¹ AND ROBERT U. NEWTON¹

¹Center for Exercise and Sport Science Research, Edith Cowan University, Perth, Australia; ²Purdue Athletics, Sports Performance Department, Purdue University, West Lafayette, Indiana; and ³Fremantle Dockers Football Club, Fremantle, Australia

ABSTRACT

Spiteri, T, Nimphius, S, Hart, NH, Specos, C, Sheppard, JM, and Newton, RU. Contribution of strength characteristics to change of direction and agility performance in female basketball athletes. *J Strength Cond Res* 28(9): 2415–2423, 2014—Research has often examined the relationship between 1 or 2 measures of strength and change of direction (COD) ability reporting inconsistent relationships to performance. These inconsistencies may be the result of the strength assessment used and the assumption that 1 measure of strength can represent all “types” of strength required during a COD task. Therefore the purpose of this study was to determine the relationship between several lower-body strength and power measures, COD, and agility performance. Twelve ($n = 12$) elite female basketball athletes completed a maximal dynamic back squat, isometric midthigh pull, eccentric and concentric only back squat, and a countermovement jump, followed by 2 COD tests (505 and T-test) and a reactive agility test. Pearson product-moment correlation and stepwise regression analysis were performed on all variables. The percentage contribution of each strength measure to an athletes total strength score was also determined. Our results demonstrated that both COD tests were significantly correlated to maximal dynamic, isometric, concentric, and eccentric strength ($r = -0.79$ to -0.89), with eccentric strength identified as the sole predictor of COD performance. Agility performance did not correlate with any measure of strength ($r = -0.08$ to -0.36), whereas lower-body power demonstrated no correlation to either agility or COD performance ($r = -0.19$ to -0.46). These findings demonstrate the importance of multiple strength components for COD ability, highlighting eccentric

strength as a deterministic factor of COD performance. Coaches should aim to develop a well-rounded strength base in athletes; ensuring eccentric strength is developed as effectively as the often-emphasized concentric or overall dynamic strength capacity.

KEY WORDS eccentric, concentric, maximal dynamic strength, power, isometric

INTRODUCTION

Change of direction (COD) and agility are common athletic maneuvers requiring athletes to possess a combination of physical, technical, and tactical attributes to evade or pursue opponents. Throughout the duration of a game, elite basketball athletes cover a distance of 991 m of high-intensity movements, executing 40–60 maximal jumps and 50–60 changes in speed and direction (1,23), emphasizing the importance of these physical attributes. The execution of efficient changes in direction both with and without the ball often determines playing performance in basketball athletes (5); therefore, athletes require a combination of perceptual-cognitive factors and strength characteristics to maintain fast directional changes and gain positional advantages during competition.

Change of direction ability involves a combination of multiple strength components (3); however, current research has often only examined the relationship between 1 or 2 measures of strength and COD performance (4,25,30). This research has identified both strong and weak correlations between lower-body strength (as measured by a maximal dynamic squat) and COD performance. An athlete's ability to shift their momentum during directional changes requires sufficient eccentric (braking), isometric (plant phase), and concentric (propulsive) strength to rapidly decelerate and reaccelerate in the new direction. As a result, clear relationships between isolated measures of strength may not be observed when the movement requires the simultaneous involvement of multiple strength components. Furthermore, during game environments, unanticipated directional

Address correspondence to Tania Spiteri, t.spiteri@ecu.edu.au.

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changes often occur in sequence, and more frequently than planned directional changes. Although previous research has observed weak relationships between strength and agility performance (20,30), the execution of 2 unanticipated directional changes may increase the muscular demands required during agility movements.

The relationship between measures of strength, COD, and agility performance is not entirely linear (29,38); however, training programs focusing on the development of lower-body strength and power (18,24,25,39) appear to produce improvements in COD performance. Furthermore, research studies examining beyond a cross-sectional relationship, investigating the relationship between changes in strength (1 repetition maximum [1RM] squat), and changes in COD performance over time have indicated moderate to strong correlations (-0.39 to -0.70) exist (17). The inconsistent relationship observed between strength and COD ability can be attributed to numerous factors including differences in the level of athlete, gender, training experience, and more importantly the strength assessment used (25). As athletes are only required to maximize their performance within the constraints and demands of their sport, COD and agility can therefore be viewed as a context-specific motor skill. It would therefore seem advantageous to use a strength assessment that replicates similar lower-body kinematics and muscle actions required when changing direction. Although only few research studies exist investigating the relationship between a multi-joint strength assessment and COD and agility performance (15,25), it can be assumed that the strength of relationship would increase when measuring multiple components of strength because of a greater level of similarity and transfer between the skill and strength assessment.

The purpose of this study was to investigate the relationship between lower-body strength components (concentric, eccentric, maximal dynamic strength, isometric, and power) and COD (505 and *T*-test) and agility performance in elite female basketball players. It was envisaged that the findings of this investigation would (a) determine the contribution of each strength component to functional COD and agility performance in elite female athletes and (b) identify which strength components best predicts COD and agility performance.

Practically, the purpose of this research is specifically aimed at improving our understanding of the specific aspects of "strength" that may be associated with COD performance lending to a greater understanding of the training prescription required to improve the underpinning physical qualities required to change in direction efficiently.

METHODS

Experimental Approach to the Problem

This study used a between subjects design to determine the relationship between various strength measures (isometric, eccentric only, concentric, dynamic, and lower-body power) and COD and agility performance. Participants were required to attend 3 testing sessions; the first consisted of a 1RM back squat assessment; the second consisted of isometric midthigh

pull, countermovement jump (CMJ), COD (505 and *T*-test) and agility assessments; and the third consisted of an eccentric and concentric back squat assessment. All testing sessions were separated by 1 week to ensure any fatigue experienced in the previous testing session did not influence the results. All testing occurred before any scheduled training sessions for that week, with a standardized 10-minute dynamic warm-up performed before any testing. Participants were also not allowed to perform any strenuous activity or lower-body resistance training within 48 hours of their assigned testing session.

Subjects

Twelve ($n = 12$) female basketball athletes (age: 24.25 ± 2.55 years; height: 177.69 ± 7.25 cm; body mass: 75.56 ± 14.55 kg) playing for the West Coast Waves Women's National Basketball League (WNBL) were recruited for this study. All athletes were recruited from the same WNBL team consisting of 3 guards, 6 forwards, and 3 centers. For inclusion into this study, participants were required to have played basketball for a minimum of 5 years and partake in a minimum of 1 competitive game(s) and 2 structured skills training sessions each week. Testing occurred after preseason training to ensure adequate fitness and minimal fatigue (as a result from in-season competitive games) for all participants before the commencement of testing. All participants were required to be injury free (of the lower limbs) at the time of testing, and report no history of major lower limb injuries such as anterior-cruciate ligament injuries or broken bones. All testing procedures were explained to the participants and written informed consent was provided. Ethics approval was obtained from the University's Human Research Ethics Committee.

Procedures

Maximal Dynamic Strength Assessment. Maximal dynamic strength was assessed using a 1RM back squat to a depth of 90° knee flexion (36). Before test commencement, participants were instructed to position their feet shoulder width apart with an unloaded Olympic bar positioned across the top of their shoulders (trapezius muscle) and lower to a half squat position (90° knee angle). Knee angle was measured by a goniometer, and an elastic band was placed around the back of the squat rack as an external guide for the required squat depth. Initially, a warm-up was performed consisting of 5–10 repetitions at 40–60% of the subjects' perceived maximum. After a 3-minute rest period, subjects then performed 3–5 repetitions at 60–80% of their perceived maximum. After another 3-minute rest period, weight was subsequently increased with subjects performing 1 repetition at the new load, resting another 3 minutes before increasing the load and performing another trial (10,36). Load was continually increased until participants could not perform the repetition with good technique (i.e., lowering to the required knee angle). A maximum of 5 attempts at any given load was permitted. The maximum load lifted is represented as a value relative to body mass. Previous test-retest reliability

coefficient following this procedure ranged between $R = 0.79$ and 0.99 (10).

Concentric and Eccentric Strength Assessment. Participants were instructed to position their feet shoulder width apart, toes slightly turned outwards, and position an Olympic bar behind their neck across the trapezius muscle. The concentric strength protocol required participants to begin seated on a box (box squat) with a knee angle of 90° , extending at the hip and knee to a standing position (32). The eccentric protocol required participants to commence from a standing position, and flex at the hip and knee to lower themselves onto a box, into a half squat position with a knee angle of 90° (36,37). Knee angle was monitored as described above in the maximal dynamic strength protocol. Familiarization trials were performed for both concentric and eccentric protocols, consisting of 5 repetitions at 40% of participants' individual 1RM. Following familiarization, participants commenced the concentric protocol first, whereby weight was added in a linear progression at 60 and 80% of maximal dynamic strength 1RM with participants performing 2–3 repetitions at each load. Once 80% of the subjects' 1RM was reached, weight

was subsequently increased with participants performing 2–3 repetitions, before increasing to a new load. Weight was continually increased until participants could not perform the repetition with good technique or could not lift the required load (13,32). A rest period of 3 minutes between each set of repetitions was provided. Once the concentric protocol was completed, participants were provided with a 10-minute rest period before their eccentric assessment. The eccentric protocol followed the same progressive loading procedure as explained for the concentric protocol with weight increased until participants could no longer controllably lower the required load to a half squat position (13). Both concentric and eccentric movements were performed to a 3-second cadence (13,26) as set by an external metronome recording. Failure to maintain the 3-second cadence during movements resulted in an unsuccessful completion of that load (13,26). The maximum load lifted both concentrically and eccentrically for each participant is presented as a value relative to body mass.

Isometric Strength Assessment. Participants performed a maximal isometric midhigh pull on a portable force plate, sampling at 600 Hz (BP12001200; AMTI, Watertown, MA, USA), with both the knee and hip angle set at approximately 140° angle (11). Participants were instructed to drive their feet (positioned shoulder width apart) into the force plate as hard and as fast as they could at the commencement of each trial. Subjects were required to perform a total of 3 trials, each trial lasting 5 seconds in duration (27), and separated by a 2-minute recovery period. The force trace for each trial was collected in Bioware software (Version 5x, Type 2812A; Kistler, Switzerland) and exported to MatLab (Version R2010a; Mathworks, Natick, MA, USA) where the average peak force of the 3 trials identified presented as a value relative to body mass ($\text{N} \cdot \text{kg}^{-1}$).

Countermovement Jump Assessment. Participants performed a CMJ on a portable force plate (400 Series Performance Plate; Fitness Technology, Adelaide, Australia) sampling at 600 Hz.

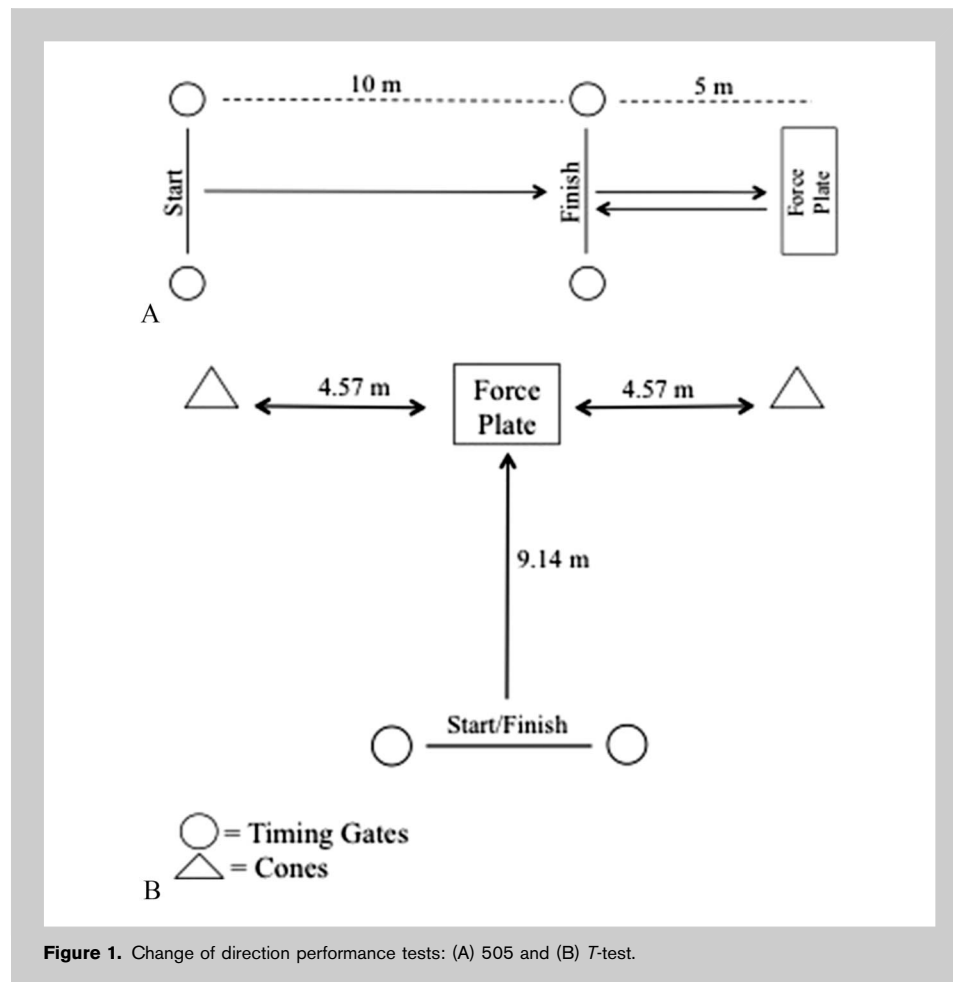


Figure 1. Change of direction performance tests: (A) 505 and (B) T-test.

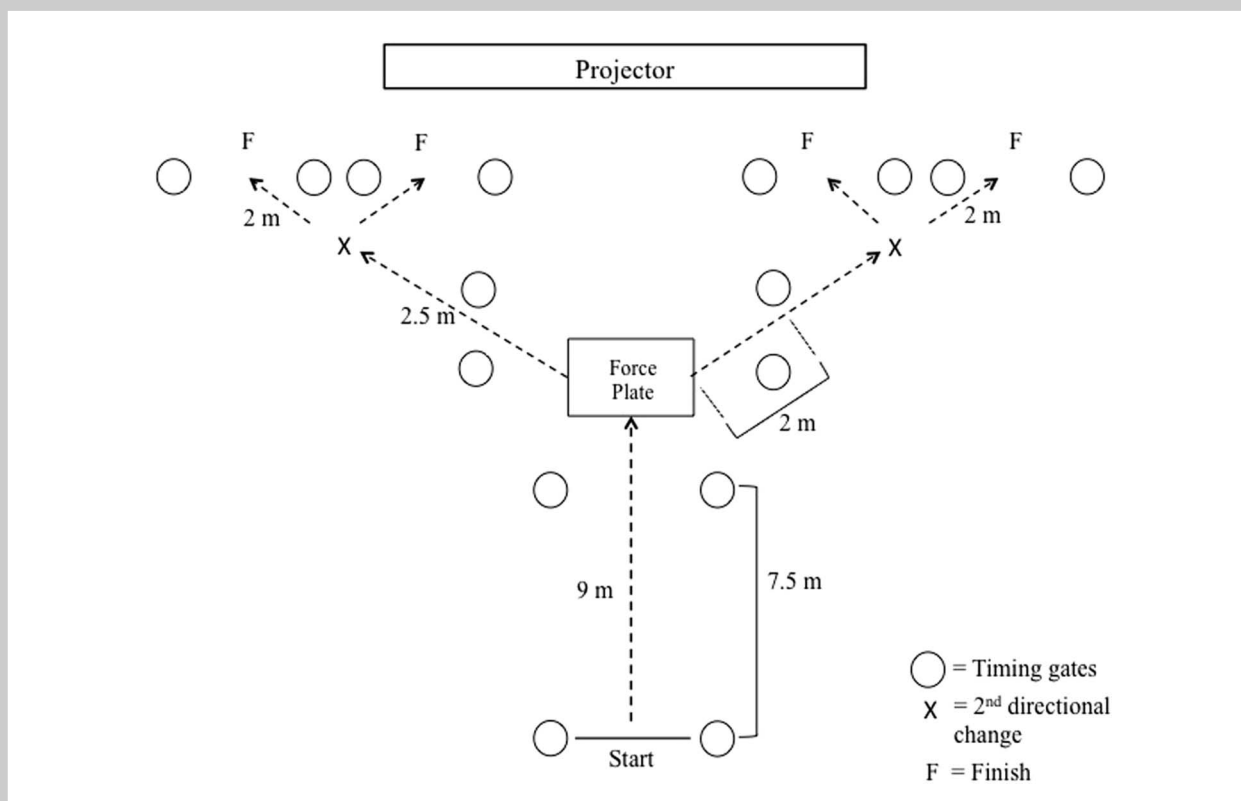


Figure 2. Layout of the reactive agility test.

Participants were instructed to stand with their feet shoulder width apart, with their hands positioned on a carbon fiber bar placed across the top of their shoulders and were instructed to jump as high as possible (21). The force trace was collected using Ballistic Measurement System Software (Version 3.4; Fitness Technology), where the average relative power ($W \cdot kg^{-1}$) was determined from 3 CMJs.

505 Protocol. Participants began standing at a set of timing gates (Speedlight Timing System; Swift Performance Equipment, Carole Park, Queensland, Australia), before sprinting 10 m through a second set of timing gates, then sprinting a further 5 m (7), before contacting their foot on a marked line, turning 180°, and finish the test by sprinting 5 m back through the timing gates (Figure 1A). Participants completed 6 trials in a randomized order; 3 changing direction with a left foot plant, and 3 changing direction with a right foot plant. Only dominant leg COD trials were used in the subsequent analysis. Limb dominance was defined as the limb that participants used as their preferred takeoff foot when performing a lay-up. The 505 COD time (s) was averaged across 3 dominant leg trials for each participant.

T-Test Protocol. Participants began standing at a set of timing gates (Speedlight Timing System; Swift Performance

Equipment), before maneuvering throughout the course performing a series of forward, backwards, and lateral movements (28), finishing by crossing back through the original timing gates (Figure 1B). Participants completed 6 trials in total; 3 initiating the lateral movement to the right first, and 3 initiating lateral movement to the left first, performed in a randomized order. Only dominant leg agility trials were used in the subsequent analysis. The average time (s) across the 3 dominant leg trials were determined for each participant.

Agility Protocol. Video clips for use within the agility test were constructed using a Sony HDD Camcorder (HDRXR550V; Sony, Sydney, Australia), recording at 200 Hz, with all footage recorded from a defensive player perspective on an indoor basketball court. The camera was positioned 16 m in front of the athlete performing the movements at a height of 1.5 m (1 athlete performed all movements in the decision-making sequence) (6,31). The athlete performing these movements had sufficient time to practice these movements before filming and has played basketball at an elite level for 14 years. The athlete was asked to run toward the camera while dribbling the ball approximately 6 m away from the camera, and execute 1 of these 8 movements (31):

1. Change direction by 45° to the left.

TABLE 1. Athletes individual maximal dynamic strength, eccentric strength, concentric strength, isometric strength, power, T-test, 505 COD and agility time with the percentage (%) contribution of each individual strength measure to athletes individual total strength and power score.*

Participant	Dynamic strength, kg · BW ⁻¹	Eccentric, kg · BW ⁻¹	Concentric, kg · BW ⁻¹	Isometric, N · BW ⁻¹	Power, W · BW ⁻¹	T-test, s	505 COD, s	Agility, s
1	1.13 (19.58)	1.43 (24.80%)	1.05 (18.27%)	1.53 (26.53)	0.62 (10.81)	12.51	2.63	5.48
2	1.11 (9.36)	1.55 (13.10)	1.03 (8.74)	1.70 (14.38)	0.78 (6.61)	9.96	2.62	5.00
3	0.82 (8.44)	0.96 (9.85)	0.76 (7.74)	0.89 (9.18)	0.57 (5.93)	12.94	3.25	5.32
4	1.04 (10.99)	1.49 (15.79)	1.10 (11.67)	1.46 (15.47)	0.56 (5.93)	10.93	2.48	5.10
5	0.82 (7.79)	0.97 (9.20)	0.67 (6.37)	0.89 (8.46)	0.68 (6.48)	12.88	2.82	4.82
6	0.89 (11.30)	1.14 (14.52)	0.76 (9.68)	1.97 (15.26)	0.43 (5.59)	11.99	2.68	4.36
7	1.13 (11.09)	1.35 (13.18)	1.06 (10.40)	1.05 (10.29)	0.67 (6.61)	11.76	2.72	4.90
8	1.24 (10.28)	1.71 (14.14)	1.24 (10.28)	1.41 (11.69)	0.70 (5.80)	11.97	2.41	4.91
9	1.64 (11.78)	2.01 (14.46)	1.49 (10.71)	2.00 (14.41)	0.71 (5.16)	9.67	2.23	4.54
10	1.03 (7.63)	1.26 (9.39)	0.95 (7.04)	1.16 (8.55)	0.74 (5.47)	13.04	3.02	4.67
11	0.58 (10.15)	0.75 (13.05)	0.50 (8.70)	0.82 (14.24)	0.39 (6.81)	11.60	2.69	5.33
Mean ± SD	1.04 ± 0.04	1.34 ± 0.34	0.96 ± 0.23	1.35 ± 0.38	0.59 ± 0.05	11.75 ± 1.15	2.69 ± 0.28	4.95 ± 0.35
% Contribution	19.70	25.39	18.18	25.56	11.17			

*BW = body weight; COD = change of direction.

2. Change direction by 45° to the right.
3. Change direction by 45° to the left and pass the ball left.
4. Change direction by 45° to the right and pass the ball right.
5. Change direction by 45° to the left and pass the ball right.
6. Change direction by 45° to the right and pass the ball left.
7. Fake right, change direction by 45° to the left, and pass the ball left.
8. Fake left, change direction by 45° to the right, and pass the ball right.

The video clips were edited using SiliconCoach Pro software (version 6.1.5.0; SiliconCOACH Ltd, Dunedin, New Zealand) so that in any clips where the ball was released; it was occluded with a black screen at 0.00 seconds of ball release. All video clips were edited to start playing 0.80 seconds before the point of occlusion (6,31). To ensure subjects could not anticipate the stimulus, a delay period ranging between 0 and 500 milliseconds was used. The videos were also programmed to trigger in a randomized order so that participants could not anticipate the correct direction of travel or action of stimulus.

Participants began standing at timing gates (Fitness Technology) on a marked line 19 m away from the projection screen and were instructed to run in a straight line toward the projected image (Figure 2). Once the participant reached the timing gates positioned 7.5 m from the starting position, the first visual stimulus (video clip) was programmed to automatically start. Participants then responded to the first video by changing direction, 45 ± 5° to the left or right moving in the same direction as the stimulus (i.e., from a defenders perspective). Once changing direction, participants then ran a further 3.5 m, triggering another set of timing gates positioned 1.5 m after the first COD, to subsequently trigger the second visual stimulus (video clip). Participants then changed direction a second time, 45 ± 5° either to the left or right in the same direction as the stimulus through the final set of timing gates to complete the test. Triggering of the video was automatically programmed through a customized code written in Kinematic Measurement System software (Version 13.0; Fitness Technology). The average total running time (s) was determined across 8 dominant leg trials where the first direction change was performed with the participant planting with their dominant leg. Reliability of this protocol was performed before testing reporting high test-retest reliability (intraclass coefficient [ICC] = 0.81, coefficient of variance [CV] = 3.3%).

Statistical Analyses

Means and SDs were calculated for all performance measures and strength assessments for all participants. The percentage contribution of each strength and power assessment to each athlete's individual total strength score (i.e., the sum of maximal dynamic strength, isometric, concentric, eccentric and power scores) was also determined for all subjects.

TABLE 2. Person correlation coefficients (*r*) between isotonic, eccentric, concentric, isometric, power, *T*-test time, 505 COD, and agility time.*

	Dynamic strength	Eccentric	Concentric	Isometric	Power	<i>T</i> -test	505 COD	Agility
Dynamic strength, kg·BW ⁻¹	1							
Eccentric, kg·BW ⁻¹	0.942†	1						
Concentric, kg·BW ⁻¹	0.953†	0.965†	1					
Isometric, N·BW ⁻¹	0.810†	0.896†	0.784†	1				
Power, W·BW ⁻¹	0.252	0.253	0.241	0.194	1			
<i>T</i> -test, s	-0.800†	-0.878†	-0.791†	-0.854†	-0.466	1		
505 COD, s	-0.804†	-0.892†	-0.791†	-0.792†	-0.166	0.810†	1	
Agility, s	-0.362	-0.271	-0.272	-0.086	-0.198	0.280	0.274	1

*BW = body weight; COD = change of direction.

†*p* ≤ 0.001.

Pearson's product-moment correlation (*r*) was performed to determine the relationships between COD (505 and *T*-test) and agility times, and strength and power measures. The strength of the correlation coefficient was described as per Hopkins (14). An *r* value between 0 to 0.30, or 0 to -0.30 was considered small; 0.31 to 0.49, or -0.31 to -0.49, moderate; 0.50 to 0.69, or -0.50 to -0.69, large; 0.70 to 0.89, or -0.70 to -0.89, very large; and 0.90 to 1, or -0.90 to -1, near perfect for predicting relationships. Multiple regression analysis was used to estimate the best predictor model of COD and agility. Coefficients of determination (*R*²) and standard error of the estimate were used to represent the goodness of the predictor models with strength and power measures as independent variables. The level of significance was set at *p* ≤ 0.05 for all the statistical analyses, unless otherwise stated. All statistical analyses were processed using the IBM SPSS statistics (Version 20.0; IBM Corporation, NY, USA).

TABLE 3. Pearson correlation coefficient (*R*) and coefficients of determination (*R*²) between predicted strength variable (eccentric strength) and change of direction (*T*-test and 505) performance times.*

	<i>R</i>	<i>R</i> ²	Adj. <i>R</i> ²	SEE	Significance
<i>T</i> -test, s	0.892	0.795	0.772	0.559	0.001
505 COD, s	0.878	0.771	0.745	0.142	0.001

*Adj. *R*² = adjusted coefficient of determination; SEE = standard error of the estimate.

RESULTS

The assessment of lower-body strength and power, COD, and agility tests for all variables are shown in Table 1. The results are displayed for each individual participant as well as the mean and *SD* for each variable measure. The percentage contribution of each strength measure (all participants combined) reveals eccentric strength and isometric strength provide the highest overall contribution (25.34 and 24.60%, respectively), followed by maximal dynamic strength (dynamic) (19.79%), concentric strength (18.34%), and power (11.94%) measurements. However, when looking at each participant's strength and power contribution to their total strength score, the contribution of each strength measurement varies.

A correlation matrix showing the Pearson correlation coefficients (*r*) between all outcome variables is presented in Table 2. Maximal dynamic strength, eccentric, concentric, and isometric strength assessments revealed a negative a significant correlation to both COD tests (*T*-test: *r* = -0.79 to -0.89, *p* = 0.001; 505: *r* = -0.79 to -0.87, *p* = 0.001), whereas no significant correlation was observed between any strength and power measure to agility performance (*r* = -0.08 to -0.36, *p* = 0.43-0.59). Performance time achieved in both the *T*-test and 505 COD assessment was also significantly correlated (*r* = 0.81, *p* = 0.001).

Stepwise multiple regression analysis revealed a statistically significant model (*p* = 0.001) with eccentric strength as the sole predictor of *T*-test and 505 COD performance accounting for 79.5 and 77.1% of the explained variance, respectively (Table 3).

DISCUSSION

This is the first study to provide data concerning the relationship of multiple strength measures to predict COD and agility performance in elite female basketball athletes. The main findings of this study were the significant negative

correlations between maximal dynamic strength, eccentric, concentric, and isometric strength and COD performance (*T*-test and 505) (Table 2), whereas no relationship to agility performance was observed. Secondly, although regression analysis revealed *T*-test and 505 performances are predominantly predicted by eccentric strength, the percent contribution of each strength measure (Table 1) to athletes overall strength score differs greater. This highlights the important notion that 1 measure of strength alone does not provide an adequate representation of an athlete's overall strength capacity, particularly when examining the relationships between strength and performance outcomes, such as time to complete COD and agility tasks.

Our results are consistent with previous research when reporting the importance of maximal dynamic strength (as measured by a 1RM) (29,35) and isometric strength for COD performance in female athletes (33). Peterson et al. (29) found a strong correlation between 1RM strength and *T*-test performance in female collegiate athletes ($r = -0.63$), whereas Nimphius et al. (25) found strong to very strong significant relationships ($r = -0.50$ to -0.75) between relative maximal dynamic strength and 505 COD performance of the dominant leg in female softball athletes. Given that COD ability is a combination of all 3 strength components (eccentric, isometric, and concentric); the development of maximal strength is crucial to create a base for other strength components (15). Additionally, as directional changes are a unilateral movement involving contact times between 0.24 and 0.40 seconds during the plant phase (2,33), possessing a sufficient level of isometric strength would also appear beneficial to maintain the required body position throughout this phase of the movement. This may also explain the slightly higher correlation of isometric strength to *T*-test performance as a greater number of directional changes and body reorientation is required to change direction compared with the 505 COD test.

Many studies have found the relationship between concentric strength and COD performance to be nonsignificant (2,38,39). In contrast to this, our findings reveal a significant ($p = 0.001$) negative correlation between concentric strength, *T*-test, and 505 COD performance ($r = -0.791$). This is similar to previous research reporting a significant relationship ($r = -0.379$ – -0.673) between several lower limb isokinetic strength tests and functional performance tests including a single leg hop assessment in female athletes (24). The inconsistent relationship between concentric strength and COD performance may be attributed to the concentric assessment used. Although a majority of previous research has assessed concentric strength through isokinetic dynamometry involving extension of the knee joint in isolation (19,24), the current study used a box squat, a multi-joint concentric action requiring the simultaneous extending of the hip, knee, and ankle. Lower-body kinematic assessment of COD performance emphasizes the coordination of both the hip and knee to improve acceleration out of directional changes (9,33).

This may explain the significantly stronger correlation to concentric strength observed in the current study, as the strength assessment replicates a similar muscular and kinematic action of the lower limbs, required when changing direction.

The importance of eccentric strength to effectively decelerate during COD movements is well documented (16,33). Previously, isokinetic tests and drop jump assessments have been used to determine an athlete's eccentric strength capacity, with research reporting significant correlations ($r = -0.52$ to 0.71) to COD performance (3,4,16,39). Our results are in accordance to these previous studies reporting a negative and significant correlation between an eccentric squat, 505 ($r = -0.892$) and *T*-test ($r = -0.878$) performance. Research examining ground reaction force and impulse production during COD movements have also identified the importance of braking capacity (eccentric strength) to improve acceleration out of directional changes (33). As eccentric strength was the sole predictor of both 505 and *T*-test performance, we can conclude that a greater emphasis of braking capacity is required when the severity of the directional change (505 test) and number of directional changes (*T*-test) increase. As increased force application during the braking phase of the movement is required to decelerate an athlete's momentum (8,33), the greater number and degree of directional change increases the eccentric muscle involvement during the movement. Therefore it appears that an athlete's ability to tolerate a greater eccentric load is a critical muscular attribute required to produce a successful COD movement.

It is often theorized that athletic movements are dependent upon the production of maximal power as a mechanism to improve performance outcomes under varying conditions (20). However, findings of the current study support those of previous research reporting nonsignificant correlations ($r = -0.35$ to 0.36) between COD tests such as shuttle runs, the 505 test and salmon runs with lower-body power production (15,20,25,38). The low relationship observed between COD performance and muscular power is somewhat surprising considering a vast majority of research has observed improvements in COD ability following training programs involving the development of power and explosive ability of the lower body (3,12,22). However, it appears that the precise measurement of lower-body power during functional movement appears to be context specific. It can be assumed that an athletes' performance during the 505 (for example) is largely dependent on the approach strategy used. Specifically, athletes would be required to decelerate their velocity and therefore momentum at greater multiples of body weight if they are entering the directional change at a higher velocity. This may explain why research has observed significant correlations ($r = -0.39$ to -0.85) between power produced during loaded CMJs and 505 COD performance time (15,25). As approach velocity was not measured in the current study, we can assume that the weak correlations observed are indicative of the method in which lower-body

power was assessed. Therefore, these findings demonstrate that power is perhaps a specific measure, which should be appropriately assessed, possibly under loaded conditions, in consideration of the particular COD test being performed.

The execution of a successful agility movement involves a combination of physical attributes and perceptual-cognitive factors. However, given that agility is considered a multidimensional skill, the relationship to lower-body strength characteristics is often considered trivial (20,30). This is supported by the findings of the current study, reporting insignificant, trivial, to small correlations between maximal dynamic strength, isometric, eccentric, concentric, and power qualities and agility performance in female basketball athletes. Previous findings have indicated that differences in the amount and degree of directional changes during most COD tests may explain why muscular strength qualities have a stronger relationship to COD compared with agility performance (30,34,39). To address this limitation, the agility test in the current study required athletes to perform two 45° directional changes, thus increasing the amount of deceleration (eccentric muscle action) and acceleration (concentric muscle action) required. When 1 directional change has been required, either in response to a human stimulus or defensive opponent (33), previous research has observed greater braking and propulsive force application compared with COD protocols. Although greater force application is a direct result of increased muscular activation during the movement, the deterministic factors of agility performance appears to be an athlete's ability to extract and identify cues to decide the appropriate movement direction. These perceptual-cognitive processes appear to reduce the significance of lower-body strength interaction to agility performance, as unlike COD movements, the cutting direction is unknown until relevant decision-making processes occur.

It should be noted that while each strength component has a different magnitude of relationship to COD performance, the contribution of each strength characteristic (Table 1) differs for each individual athlete. Although isometric and eccentric strength account for the largest overall percentage for the group, the contribution of concentric maximal dynamic strength and muscular power to an athlete's overall strength score varied between individuals. This has significant practical implications in the way athletic performance and strength capacities are currently assessed. Often studies perform 1 or 2 measures of strength to determine an athlete's overall strength capacity and this limits practical relevance to performance outcomes as a result. Therefore, by only obtaining 1 measure of strength, we may be inaccurately categorizing athletes in terms of their strength capacity to perform performance tasks such as COD.

In conclusion, the current study demonstrates that strong correlations exist between measures of lower-body strength and COD performance in elite female basketball players. Although these findings are limited to a small ($n = 12$)

population of elite female basketball players with similar training backgrounds and playing experience, these findings can potentially translate to other field-based and court-based team sports where directional changes commonly occur. Specifically, the significant correlation between eccentric, concentric, and isometric strength highlight the importance to train and develop these strength qualities to achieve more efficient COD performance. Although limited relationship to agility was observed, the foundation of movement still requires these strength components and while they may not share a prominent relationship to agility performance outcomes they are still required to execute fast and rapid movement during reactive settings.

PRACTICAL APPLICATIONS

The results of this study demonstrate that a strong correlation exists between lower-body strength (maximal dynamic strength, isometric, concentric and eccentric strength) and COD performance in female athletes. Although no relationship was observed between any strength measure and agility performance, the development of these physical attributes cannot be ignored when training field- or court-based athletes as these strength qualities provide the foundation for movements executed during game environments. Coaches should aim to develop a well-rounded strength base or capacity in athletes, ensuring their eccentric strength capacity is developed as effectively as the often-emphasized concentric or overall dynamic strength capacity.

It would be hypothesized that improving the eccentric strength of an athlete directly improves the ability to tolerate braking loads or braking capacity required to produce efficient changes in direction, particularly when multiple directional changes, or greater degree of directional change is involved. This approach to strength development may then have a greater chance to translate strength improvements to performance improvements in COD ability. Exercises such as squats, power clean, drop landings, plyometrics, or progressive drill emphasizing the eccentric phase will enable greater opportunity to translate that strength with mechanical specificity into effective COD performance.

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